RT vs non-RT task threads - choreography-mqueue

Student and Welch t-tests on the per-iteration timings of perf.csv

Experiments compared

RT test_single_core_1 (fifo prio 50,60,70)

non-RT test_single_core_nort_2 (normal nice -8,-4,-1)

Method

99 RT and 99 non-RT iterations per task (first 1 iteration(s) dropped as warm-up); one test per task and metric.

Two-sided, unpaired tests; p-values Holm-corrected within each comparison; significance level alpha = 0.05.

Welch's test is the reference (it does not assume equal variances); Student's is reported for comparison.

Key results: end-to-end response time, test_single_core_1 vs test_single_core_nort_2

Task 1: RT 86.5 ms vs non-RT 86.8 ms -> non-RT 0.25 ms slower (large effect), Welch p < 0.001

Task 2: RT 259.7 ms vs non-RT 183.8 ms -> non-RT 75.97 ms faster (large effect), Welch p < 0.001

Task 3: RT 86.6 ms vs non-RT 209.0 ms -> non-RT 122.32 ms slower (large effect), Welch p < 0.001

Task 4: RT 153.1 ms vs non-RT 189.3 ms -> non-RT 36.22 ms slower (large effect), Welch p < 0.001

Task 5: RT 86.9 ms vs non-RT 136.8 ms -> non-RT 49.92 ms slower (large effect), Welch p < 0.001

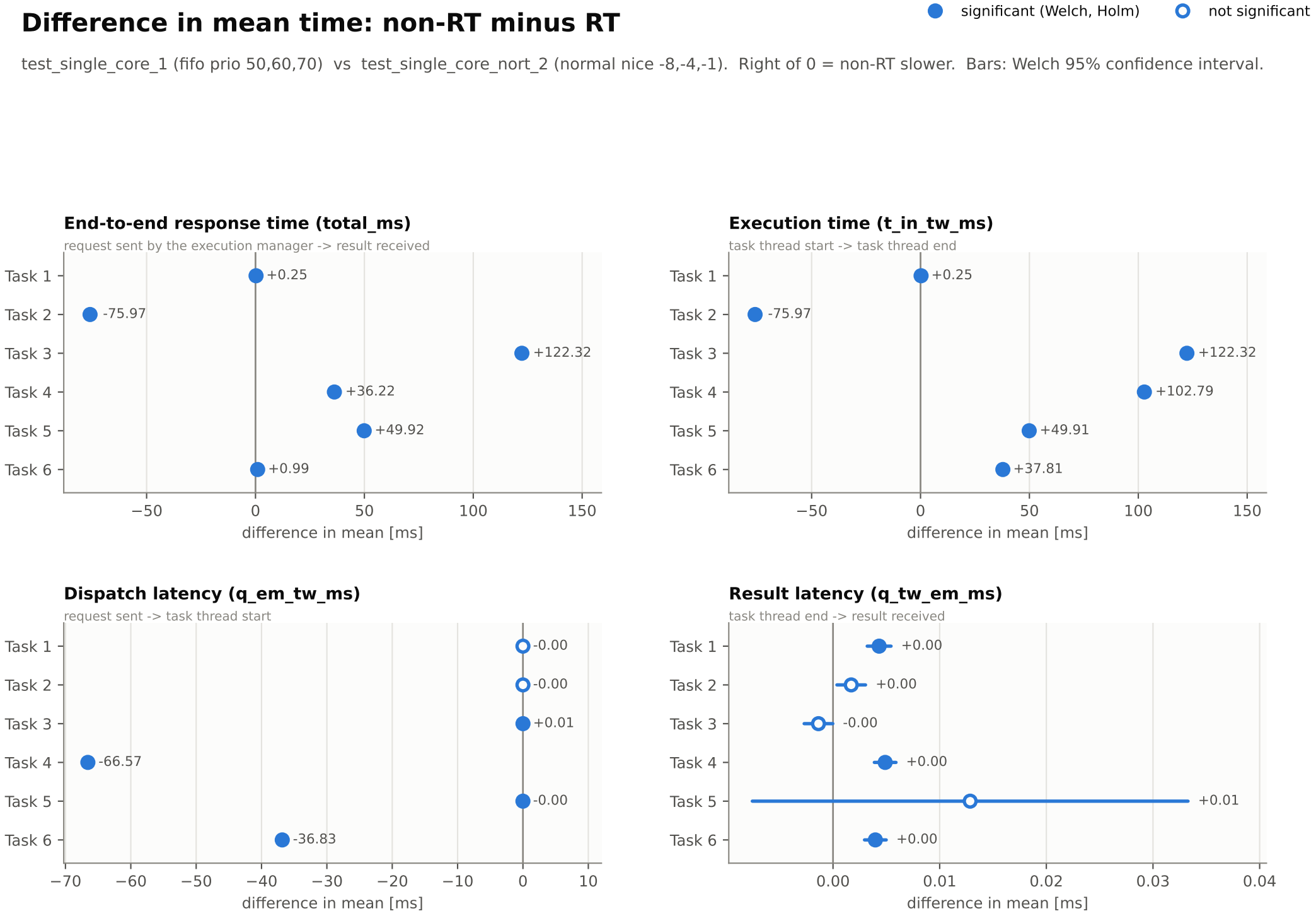
Task 6: RT 123.9 ms vs non-RT 124.9 ms -> non-RT 0.99 ms slower (large effect), Welch p < 0.001

Overall: 19 of 24 task/metric tests significant (Welch); Student and Welch disagree on 0.

Note: with ~100 samples per side, even tiny differences are significant. Judge importance by the difference in ms and the effect size, not by the p-value.

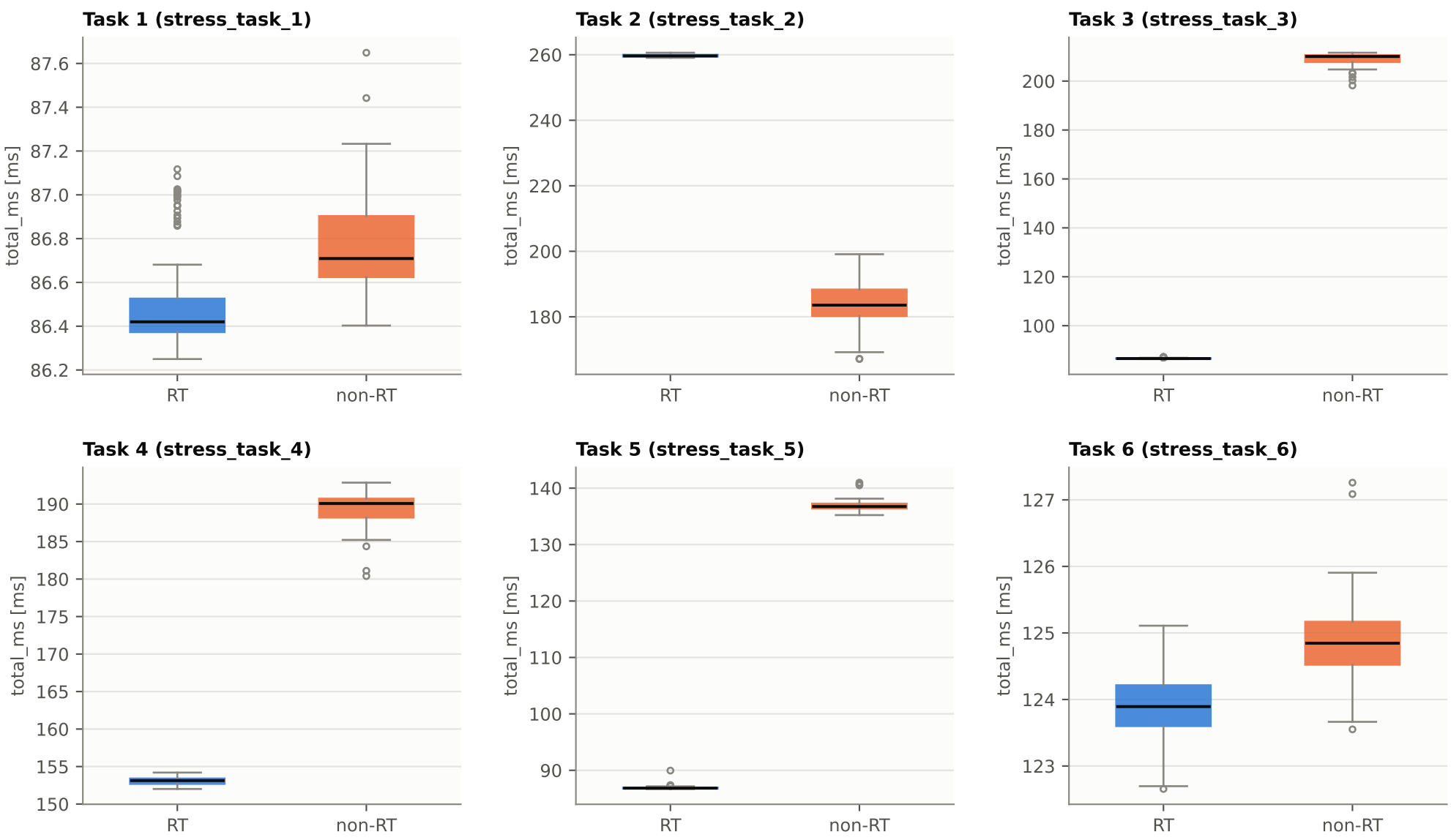
Difference in mean time: non-RT minus RT

test_single_core_1 (fifo prio 50,60,70) vs test_single_core_nort_2 (normal nice -8,-4,-1). Right of 0 = non-RT slower. Bars: Welch 95% confidence interval.



Distribution of end-to-end response time per task

test_single_core_1 (fifo prio 50,60,70) vs test_single_core_nort_2 (normal nice -8,-4,-1).
Box: 25th-75th percentile, line: median, whiskers: 1.5 IQR, dots: outliers. Each panel has its own scale.



All results

test_single_core_1 (fifo prio 50,60,70) vs test_single_core_nort_2 (normal nice -8,-4,-1). Means in ms (± standard deviation); p-values Holm-corrected.

Task	Metric	RT mean ± sd	non-RT mean ± sd	Difference [95% CI]	Student p	Welch p	Hedges' g	Verdict (Welch)
1	total_ms	86.513 ± 0.236	86.768 ± 0.225	+0.254 [+0.190, +0.319]	< 0.001	< 0.001	1.10 (large)	non-RT 0.25 ms slower (large effect)
1	t_in_tw_ms	86.419 ± 0.234	86.670 ± 0.224	+0.252 [+0.188, +0.316]	< 0.001	< 0.001	1.10 (large)	non-RT 0.25 ms slower (large effect)
1	q_em_tw_ms	0.088 ± 0.006	0.086 ± 0.006	-0.002 [-0.004, -0.000]	0.085	0.085	-0.33 (small)	no evidence of a difference
1	q_tw_em_ms	0.007 ± 0.003	0.011 ± 0.005	+0.004 [+0.003, +0.005]	< 0.001	< 0.001	1.08 (large)	non-RT 0.00 ms slower (large effect)
2	total_ms	259.736 ± 0.373	183.770 ± 6.397	-75.967 [-77.245, -74.689]	< 0.001	< 0.001	-16.70 (large)	non-RT 75.97 ms faster (large effect)
2	t_in_tw_ms	259.642 ± 0.370	183.676 ± 6.398	-75.966 [-77.244, -74.688]	< 0.001	< 0.001	-16.70 (large)	non-RT 75.97 ms faster (large effect)
2	q_em_tw_ms	0.085 ± 0.008	0.083 ± 0.006	-0.002 [-0.004, +0.000]	0.140	0.140	-0.27 (small)	no evidence of a difference
2	q_tw_em_ms	0.009 ± 0.005	0.011 ± 0.004	+0.002 [+0.000, +0.003]	0.064	0.065	0.36 (small)	no evidence of a difference
3	total_ms	86.637 ± 0.202	208.962 ± 2.569	+122.325 [+121.811, +122.839	< 0.001	< 0.001	66.87 (large)	non-RT 122.32 ms slower (large effect)
3	t_in_tw_ms	86.561 ± 0.201	208.882 ± 2.570	+122.321 [+121.807, +122.835	< 0.001	< 0.001	66.84 (large)	non-RT 122.32 ms slower (large effect)
3	q_em_tw_ms	0.066 ± 0.005	0.071 ± 0.008	+0.005 [+0.004, +0.007]	< 0.001	< 0.001	0.84 (large)	non-RT 0.01 ms slower (large effect)
3	q_tw_em_ms	0.010 ± 0.005	0.009 ± 0.004	-0.001 [-0.003, -0.000]	0.140	0.140	-0.28 (small)	no evidence of a difference
4	total_ms	153.102 ± 0.482	189.327 ± 2.052	+36.225 [+35.805, +36.644]	< 0.001	< 0.001	24.21 (large)	non-RT 36.22 ms slower (large effect)
4	t_in_tw_ms	86.445 ± 0.201	189.236 ± 2.053	+102.791 [+102.379, +103.202	< 0.001	< 0.001	70.20 (large)	non-RT 102.79 ms slower (large effect)
4	q_em_tw_ms	66.652 ± 0.446	0.081 ± 0.005	-66.571 [-66.660, -66.482]	< 0.001	< 0.001	-210.29 (large)	non-RT 66.57 ms faster (large effect)
4	q_tw_em_ms	0.005 ± 0.001	0.010 ± 0.005	+0.005 [+0.004, +0.006]	< 0.001	< 0.001	1.35 (large)	non-RT 0.00 ms slower (large effect)
5	total_ms	86.909 ± 0.345	136.825 ± 0.916	+49.916 [+49.721, +50.111]	< 0.001	< 0.001	71.85 (large)	non-RT 49.92 ms slower (large effect)
5	t_in_tw_ms	86.817 ± 0.343	136.723 ± 0.914	+49.906 [+49.712, +50.101]	< 0.001	< 0.001	72.01 (large)	non-RT 49.91 ms slower (large effect)
5	q_em_tw_ms	0.083 ± 0.007	0.080 ± 0.009	-0.003 [-0.005, -0.001]	0.030	0.030	-0.40 (small)	non-RT 0.00 ms faster (small effect)
5	q_tw_em_ms	0.009 ± 0.006	0.022 ± 0.102	+0.013 [-0.008, +0.033]	0.213	0.215	0.18 (negligible)	no evidence of a difference
6	total_ms	123.881 ± 0.458	124.868 ± 0.566	+0.987 [+0.842, +1.131]	< 0.001	< 0.001	1.91 (large)	non-RT 0.99 ms slower (large effect)
6	t_in_tw_ms	86.973 ± 0.126	124.783 ± 0.561	+37.810 [+37.696, +37.925]	< 0.001	< 0.001	92.67 (large)	non-RT 37.81 ms slower (large effect)
6	q_em_tw_ms	36.903 ± 0.421	0.075 ± 0.022	-36.827 [-36.912, -36.743]	< 0.001	< 0.001	-122.97 (large)	non-RT 36.83 ms faster (large effect)
6	q_tw_em_ms	0.006 ± 0.002	0.010 ± 0.005	+0.004 [+0.003, +0.005]	< 0.001	< 0.001	1.09 (large)	non-RT 0.00 ms slower (large effect)

How to read this report

Metrics

total_ms	End-to-end response time: request sent by the execution manager -> result received
t_in_tw_ms	Execution time: task thread start -> task thread end
q_em_tw_ms	Dispatch latency: request sent -> task thread start
q_tw_em_ms	Result latency: task thread end -> result received
	total_ms = q_em_tw_ms + t_in_tw_ms + q_tw_em_ms. Use total_ms for the overall effect: waiting can move between the parts.
Difference	mean(non-RT) - mean(RT). Positive = non-RT slower, negative = non-RT faster.
95% CI	range that contains the true difference with 95% confidence (Welch). If it does not cross 0, the difference is significant.
p-value	probability of a difference at least this large if the true means were equal. < 0.05 = significant. '< 0.001' means practically zero.
Holm	correction for running many tests at once, so that 'significant' results are not found by chance.
Student vs Welch	Student assumes equal variances, Welch does not (reference). A much smaller Student p signals that the variances differ.
Hedges' g	difference in units of standard deviation: < 0.2 negligible, < 0.5 small, < 0.8 medium, otherwise large.
Caveats	a non-significant result is not proof of equality; the t-test compares means only - check the distributions page for jitter and outliers.